

|                |                                                                                                                                                                                                                                                                                                                |                  |                   |                    |             |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------|--------------------|-------------|
| Programme      | BSc Statistics                                                                                                                                                                                                                                                                                                 |                  |                   |                    |             |
| Course Code    | STA1MN106 (P)                                                                                                                                                                                                                                                                                                  |                  |                   |                    |             |
| Course Title   | Introductory statistics with JASP                                                                                                                                                                                                                                                                              |                  |                   |                    |             |
| Type of Course | Minor                                                                                                                                                                                                                                                                                                          |                  |                   |                    |             |
| Semester       | I                                                                                                                                                                                                                                                                                                              |                  |                   |                    |             |
| Academic Level | 100 - 199                                                                                                                                                                                                                                                                                                      |                  |                   |                    |             |
| Course Details | Credit                                                                                                                                                                                                                                                                                                         | Lecture per week | Tutorial per week | Practical per week | Total Hours |
|                | 4                                                                                                                                                                                                                                                                                                              | 3                | -                 | 2                  | 75          |
| Pre-requisites | Basic mathematical and computer skills. Basic knowledge of probability theory.                                                                                                                                                                                                                                 |                  |                   |                    |             |
| Course Summary | Introduce statistical concepts with JASP software, covering data collection, questionnaire types, measurement scales, and graphical representation, while familiarizing students with installation, file manipulation, and descriptive statistics application, preparing for practical analysis in Psychology. |                  |                   |                    |             |

#### Course Outcomes (CO):

| CO  | CO Statement                                                                                                                                       | Cognitive Level* | Knowledge Category# | Evaluation Tools used           |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|---------------------------------|
| CO1 | Understand data collection methods and questionnaire types and critically evaluate ethical implications of statistical methods aligning with human | U                | C                   | Instructor-created exams / Quiz |

|                                                                                                                                                                                                |                                                                                                                                             |    |   |                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|----|---|-----------------------------------------------------------------------|
|                                                                                                                                                                                                | values.                                                                                                                                     |    |   |                                                                       |
| CO2                                                                                                                                                                                            | Identify the differences between primary data and secondary data                                                                            | U  | C | Practical Assignment / Observation of Practical Skills                |
| CO3                                                                                                                                                                                            | Assess scales of measurement, reliability, and validity and analyze data to help entrepreneurial decisions using critical thinking skills.. | U  | C | Seminar Presentation / Group Tutorial Work/ Instructor-created exams  |
| CO4                                                                                                                                                                                            | Design survey questions that minimize bias and encourage accurate response.                                                                 | U  | C | Instructor-created exams / Home Assignments/ Instructor-created exams |
| CO5                                                                                                                                                                                            | Formulate and represent frequency distributions graphically.                                                                                | U  | F | One Minute Reflection Writing assignments/ Instructor-created exams   |
| CO6                                                                                                                                                                                            | Master JASP software for descriptive statistics.                                                                                            | Ap | P | Viva Voce/ Instructor-created exams                                   |
| * - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)<br># - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M) |                                                                                                                                             |    |   |                                                                       |

### Detailed Syllabus:

| Module | Unit                                   | Content                            | Hrs<br>(45+30) | Marks     |
|--------|----------------------------------------|------------------------------------|----------------|-----------|
| I      | <b>Organizing a Statistical Survey</b> |                                    | <b>10</b>      | <b>15</b> |
|        | 1                                      | Statistical Survey—An Introduction | 2              |           |
|        | 2                                      | Planning the Survey                | 1              |           |
|        | 3                                      | Specification of the Purpose       | 1              |           |
|        | 4                                      | Scope of the Survey                | 1              |           |
|        | 5                                      | Sources of Data                    | 2              |           |
|        | 6                                      | Methods of collecting primary data | 2              |           |

|            |                                                                                                                                                        |                                                        |           |           |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------|-----------|
|            | 7                                                                                                                                                      | Drafting the questionnaire                             | 1         |           |
|            | Unit 1: 2.2 Ref[1]<br>Unit 2: 2.3 Ref[1]<br>Unit 3: 2.3 Ref[1]<br>Unit 4: 2.3 Ref[1]<br>Unit 5: 2.5 Ref[1]<br>Unit 6: 3.3 Ref[1]<br>Unit 7: 3.8 Ref[1] |                                                        |           |           |
| <b>II</b>  | <b>An introduction to Research Design</b>                                                                                                              |                                                        | <b>9</b>  | <b>20</b> |
|            | 6                                                                                                                                                      | Introduction of Psychological measurement and variable | 2         |           |
|            | 7                                                                                                                                                      | Scales of measurement                                  | 2         |           |
|            | 8                                                                                                                                                      | Assessing the reliability of measurement               | 3         |           |
|            | 9                                                                                                                                                      | Assessing validity of a study                          | 2         |           |
|            | Unit 6: 2.1 Ref[1]<br>Unit 7: 2.2 Ref[1]<br>Unit 8: 2.3 Ref[1]<br>Unit 9:2.6 Ref[1]                                                                    |                                                        |           |           |
|            | <b>Graphical Representation</b>                                                                                                                        |                                                        | <b>15</b> | <b>20</b> |
| <b>III</b> | 9                                                                                                                                                      | Graphical representation of a Frequency Distribution   | 2         |           |
|            | 10                                                                                                                                                     | Histogram                                              | 1         |           |
|            | 11                                                                                                                                                     | Frequency Polygon                                      | 1         |           |
|            | 12                                                                                                                                                     | Ogives                                                 | 3         |           |
|            | 13                                                                                                                                                     | Smoothed frequency curve                               | 2         |           |
|            | 14                                                                                                                                                     | Technique of Constructing Graphs                       | 2         |           |
|            | 15                                                                                                                                                     | Graphs of Time Series or Line Graphs                   | 2         |           |

|           |                                                                                                                                                                                                                         |                                                               |           |           |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-----------|-----------|
|           | 16                                                                                                                                                                                                                      | Range Chart                                                   | 1         |           |
|           | 17                                                                                                                                                                                                                      | Band Graph                                                    | 1         |           |
|           | Unit 9: 2.2 Ref[2]<br>Unit 10: 2.2.1 Ref[2]<br>Unit 11: 2.2.2 Ref[2]<br>Unit 12: 2.11.1 Ref[2]<br>Unit 13: 6.40 Ref[1]<br>Unit 14: 6.24 Ref[1]<br>Unit 15: 6.24 Ref[1]<br>Unit 16:: 6.29 Ref[1]<br>Unit 17: 6.30 Ref[1] |                                                               |           |           |
| <b>IV</b> | <b>An Introduction to JASP</b>                                                                                                                                                                                          |                                                               | <b>11</b> | <b>15</b> |
|           | 18                                                                                                                                                                                                                      | Installing JASP                                               | 1         |           |
|           | 19                                                                                                                                                                                                                      | Loading data in JASP                                          | 1         |           |
|           | 20                                                                                                                                                                                                                      | Changing data from one measurement scale to another           | 1         |           |
|           | 21                                                                                                                                                                                                                      | Calculating Mean, Median and Mode in JASP                     | 4         |           |
|           | 22                                                                                                                                                                                                                      | Calculating Range, standard deviation and variance using JASP | 4         |           |
|           | Sections from References:<br>Unit 18: 3.1 Ref[3]<br>Unit 19: 3.3 Ref[3]<br>Unit 20: 3.5 Ref[3]<br>Unit 21: 4.1 Ref[3]<br>Unit 22: 4.2 Ref[3]                                                                            |                                                               |           |           |
| <b>V</b>  | <b>PRACTICUM</b>                                                                                                                                                                                                        |                                                               | <b>30</b> |           |

|  |                                                                                                                                                                                                                                                                              |                            |  |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--|--|
|  | Do practice problems using JASP software from any 5 units of the given list and one additional problem decided by the teacher-in-charge, related to the content of the course. Other units listed here may be used as demonstrations of the concepts taught in the course.   |                            |  |  |
|  | 1                                                                                                                                                                                                                                                                            | Standard scores in JASP    |  |  |
|  | 2                                                                                                                                                                                                                                                                            | Saving image files         |  |  |
|  | 3                                                                                                                                                                                                                                                                            | Histogram                  |  |  |
|  | 4                                                                                                                                                                                                                                                                            | Box plots                  |  |  |
|  | 5                                                                                                                                                                                                                                                                            | Drawing multiple box plots |  |  |
|  | 6                                                                                                                                                                                                                                                                            | Examples on Nominal scale  |  |  |
|  | 7                                                                                                                                                                                                                                                                            | Examples on ordinal scale  |  |  |
|  | 8                                                                                                                                                                                                                                                                            | Examples on Interval scale |  |  |
|  | 9                                                                                                                                                                                                                                                                            | Examples on Ratio scale    |  |  |
|  | <p>Sections from References:</p> <p>Unit 1: 4.5 Ref[3]</p> <p>Unit 2: 5.3 Ref[3]</p> <p>Unit 3: 5.1 Ref[3]</p> <p>Unit 4:5.2 Ref[3]</p> <p>Unit 5:5.2.2 Ref[3]</p> <p>Unit 6:2.2.1Ref[3]</p> <p>Unit7:2.2.2 Ref[3]</p> <p>Unit 8:2.2.3 Ref[3]</p> <p>Unit 9:2.2.4 Ref[3]</p> |                            |  |  |

**Books and References:**

1. S.P Gupta (2021), Statistical Methods 46 th Edition
2. Gupta, S.C. and Kapoor, V.K. (1997) Fundamentals of Mathematical Statistics. Sultan Chand and Sons, New Delhi
3. Navarro, D.J., Foxcroft, D.R., & Faulkenberry, T.J. (2019). Learning Statistics with JASP: A Tutorial for Psychology Students and Other Beginners. (Version ).

**Mapping of COs with PSOs and POs :**

|         | PSO<br>1 | PSO<br>2 | PSO<br>3 | PSO4 | PSO<br>5 | PSO6 | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 |
|---------|----------|----------|----------|------|----------|------|-----|-----|-----|-----|-----|-----|
| CO<br>1 | 3        | 3        | -        | -    | -        | 2    | 1   | 2   | -   | -   | -   | 3   |
| CO<br>2 | -        | -        | -        | -    | -        | 3    | 2   | 2   | -   | -   | -   | -   |
| CO<br>3 | -        | 2        | -        | 3    | 2        | -    | 2   | 2   | 2   | -   | 3   | -   |
| CO<br>4 | -        | -        | -        | -    | 3        | -    | 3   | 1   | 2   | -   | -   | -   |
| CO<br>5 | 3        | -        | -        | -    | -        | -    | 2   | 2   | -   | -   | -   | -   |
| CO<br>6 | -        | -        | 3        | -    | -        | 3    | 1   | 1   | -   | 3   | 2   | -   |

**Correlation Levels:**

| Level | Correlation        |
|-------|--------------------|
| -     | Nil                |
| 1     | Slightly / Low     |
| 2     | Moderate / Medium  |
| 3     | Substantial / High |

**Assessment Rubrics:**

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

**Mapping of COs to Assessment Rubrics :**

|      | Internal Exam | Assignm<br>ent | Project<br>Evaluation | End Semester<br>Examinations |
|------|---------------|----------------|-----------------------|------------------------------|
| CO 1 | ✓             |                |                       | ✓                            |
| CO 2 | ✓             | ✓              |                       | ✓                            |
| CO 3 | ✓             |                |                       | ✓                            |
| CO 4 | ✓             | ✓              |                       | ✓                            |
| CO 5 |               | ✓              |                       | ✓                            |
| CO 6 | ✓             |                |                       |                              |